

Experiment 05: Interfacing Soil Moisture Sensor with Arduino UNO

Analog Resistive Sensing & Threshold Actuation — Program Source Code & Experimental Output

1. Arduino C++ Source Code

Listing 5.1: Microcontroller Firmware for Soil Moisture Telemetry and Threshold Actuation.

```
1 /* Experiment 05: Soil Moisture Sensor Interfacing with Arduino UNO
2  * Hardware Pin Connections: HW-103 Module A0 -> Pin A0 | Alert LED -> Pin 13 | VCC/GND -> 5V/GND
3  * Operating Logic: ADC Value < 500 indicates DRY soil (Triggers Pin 13 LED ON) */
4 const int sensorPin = A0; // Soil moisture sensor connected to Pin A0
5 const int ledPin = 13;    // Status LED connected to digital pin 13
6 int sensorValue = 0;      // Variable to store raw analog sensor value
7 void setup() {
8   pinMode(ledPin, OUTPUT); // Configure Pin 13 as digital output driver
9   Serial.begin(9600);      // Initialize UART Serial Interface at 9600 Baud
10 }
11 void loop() {
12   sensorValue = analogRead(sensorPin); // Acquire 10-bit analog voltage from sensor
13   Serial.print("Soil moisture Level: ");
14   Serial.println(sensorValue);
15   // Closed-loop threshold decision: Turn ON indicator if soil is dry
16   if (sensorValue < 500) {
17     digitalWrite(ledPin, HIGH); // Dry soil condition: Alert / Pump ON
18   } else {
19     digitalWrite(ledPin, LOW);  // Sufficient moisture: LED OFF
20   }
21   delay(1000); // 1-second sampling period
22 }
```

2. Experimental Output, Serial Monitor Telemetry & Plotter Response

A. Soil Moisture Calibration & Observation Log:

#	Soil Medium Condition	ADC (0–1023)	Voltage	Indicator LED 13 Status
1	Ambient Air (Open Circuit)	33	0.16 V	ON (Dry Soil Alarm Triggered)
2	Dry Soil Specimen	29	0.14 V	ON (Dry Soil Alarm Triggered)
3	Dry Soil Specimen (Replicate)	35	0.17 V	ON (Dry Soil Alarm Triggered)
4	Moderately Moistened Soil	560	2.74 V	OFF (Moisture Setpoint Satisfied)
5	Water-Saturated Soil	820	4.01 V	OFF (Maximum Conductivity)

B. Serial Monitor Stream (9600 Baud):

```
Soil moisture Level: 33
Soil moisture Level: 29
Soil moisture Level: 35
Soil moisture Level: 38
```

C. Arduino Serial Plotter Waveform Response:

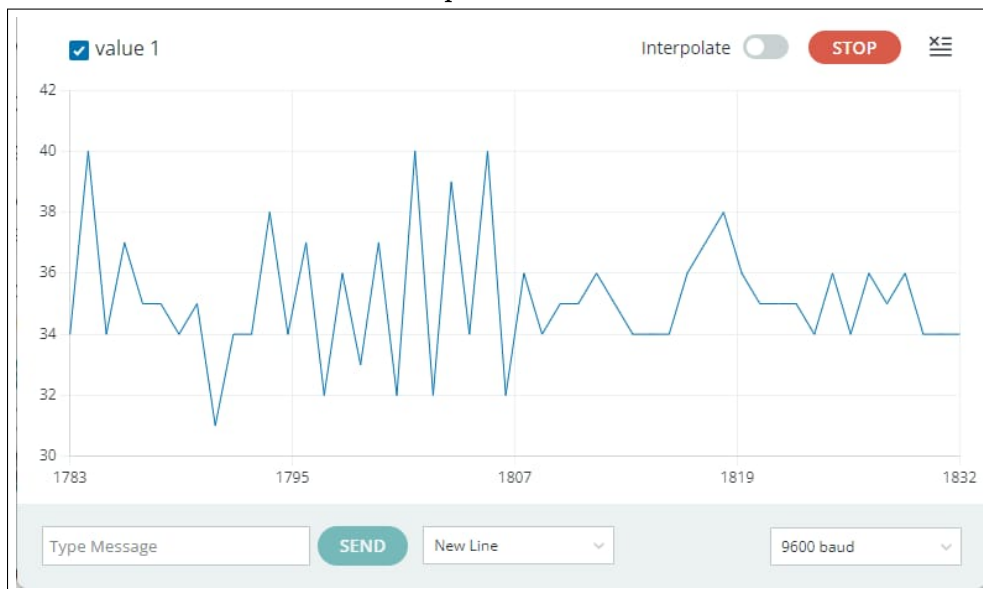


Figure 5.1: Arduino Serial Plotter waveform during dry soil baseline testing ($31 \leq \text{ADC} \leq 40$).